MATERIALS AND MANUFACTURING PROCESSES
2022, VOL. 37, NO. 4, 359–371
https://doi.org/10.1080/10426914.2021.2016816

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Wood mechanical densification: a review on processing

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ABSTRACT

As a renewable and low-price material, wood has been used extensively in the construction, furniture and other fields. However, the mechanical strength of most wood species is moderate or low, which cannot meet the need of some advanced engineering structures and applications. High-strength wood resources are scarce and usually expensive. Therefore, it is of great significance for economic and sustainable development to improve mechanical properties of medium or low-density wood. Mechanical densification process is considered an effective method to improve wood mechanical properties. Compared with chemical methods, it has great advantages of preserving the eco-friendly characteristics of nature wood and environmental protection. Many researches and technologies of wood mechanical densification have been reported and attempted. This study summarized, classified and compared the different mechanical densification processes according to their commonality and individuality. The characteristics and disadvantages of different processes were also discussed and analyzed with a purpose of inspiring researchers and entrepreneurs to ameliorate the technologies.

ARTICLE HISTORY
Received 20 August 2021
Accepted 28 November 2021

KEYWORDS
Wood; densification; heat treatment; steam treatment; thermo-hydro-mechanical densification; thermo-mechanical densification; non-adding steam densification; adding steam densification; processing; high strength wood; high density wood

Introduction

Compared with steel, concrete, plastic and other materials, wood not only has great strength and low price, but also shows attractive characteristics such as environmentally friendly and renewable. It has become an important engineering material and has played an important role in the field of construction, furniture and so on.[1] However, the mechanical strength of most wood species is moderate or low. Many applications with high strength requirements still rely on some synthetic materials or scarce high-strength wood species. With the long-term consumption of high-quality and high-density wood, as well as the policy of restriction and prohibition of logging in recent years, resources have become increasingly scarce.[2] Therefore, low or medium-strength woods with fast growth and short rotation have gained increasing attention. If their strength can be improved artificially, the application of low or medium-strength wood could be expanded significantly besides substituting for scarce and expensive high-strength wood. It is common knowledge that wood strength is positively correlated with its density. Hence, many treatments for increasing wood density (densification) have been attempted and developed in order to improve wood strength.[3] As far back as the 1930s, densified wood has been used to replace metal materials for some military applications to prevent detection by radar. Anchors and shuttles for textile made of densified wood were also reported.[4] Recently, we reported a kind of engineered wood flooring for heavy load support, the surface layer of which is densified sugar maple (*Acer saccharum*).[5]

At present, wood densification can be achieved by mechanical compression to reduce void volume of wood,[6] by impregnating wood pore space with a liquid chemical substance[7–9] or by a combination of mechanical and chemical methods,[10–14] Compared with mechanical compression, chemical impregnation not only destroys the ecological and recyclable characteristics of natural wood, but also easily causes environmental pollution, and the cost is usually high.[15,16] This review will focus on mechanical densification processes.

Mechanical densification is to change the internal microstructure, and reduce the void volume of wood by load applied on the nature wood to increase its density and achieve the purpose of improving wood mechanical properties. Mechanical densification has been investigated more than a century. In 1900, Sears applied for the first patent on densified wood product, followed by Walch and Watts, Oleheimer, Brossmannes, Esselen, and Olson in 1920s and 1930s. From 1930 to 1950, some trade names of compressed wood were used, such as Staypak, Lignifol, and Lignostone.[17,18] However, these densified wood products have poor dimension stability. They are susceptible to moisture and are prone to recover their original shapes almost completely when re-moistened and heated.[19] Since the end of the 20th century, wood industry has developed rapidly, researchers have carried out new explorations and discoveries based on the previous compression technologies, and the densification process has diversified. Due to the variety of mechanical densification processes, and the complexity of the intersections and the differentiations between them, it is difficult to classify. This paper will summarize, classify and compare the different mechanical wood densification processes according to their commonality and

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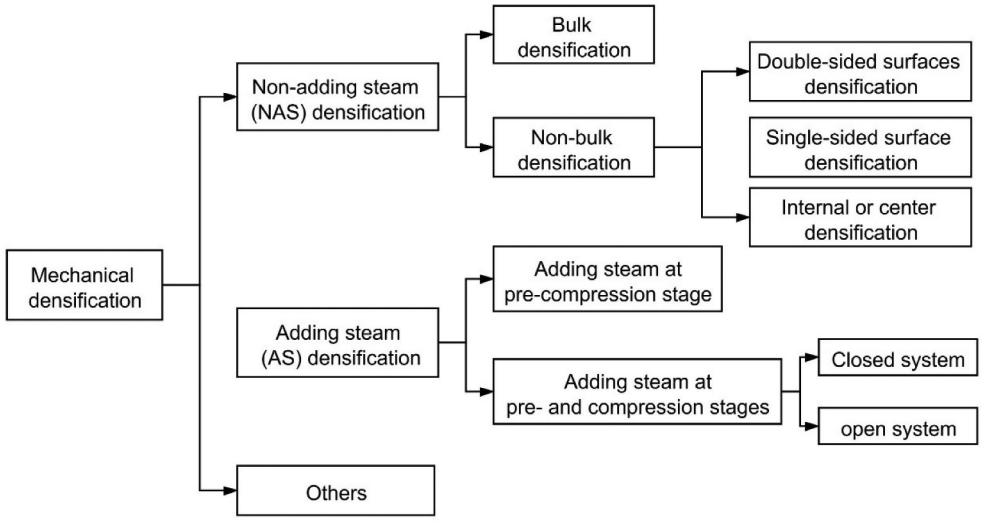

Figure 1. The classification of wood mechanical densification process.

individuality, introduce their process characteristics and the properties of densified wood, with a purpose of inspiring researchers and entrepreneurs to ameliorate the technology.

Densification processes

Wood can be compressed along the tangential, longitudinal and radial directions. However, if compressed along the tangential direction, the latewood of the annual rings would be buckled and pressed into waves shape. Longitudinal compression would produce folds and lead to considerable damages at the micro-scale.[20] Therefore, the compression direction of wood densification is usually along the radial direction, which avoids fracture during compression.[21] In this paper, only the densification process along radial direction will be discussed.

With the long-term development of the densification technology, various mechanical densification processes have been studied and applied. Because of the complex relationship and difference between them, there has no clear definition or classification of mechanical densification processes. Many studies have described the process as thermo-hydro-mechanical (THM) densification or thermo-mechanical (TM) densification according to whether the moisture content of wood is controlled in the process. However, due to the diversity of methods affecting wood moisture content and the non-uniqueness of moisture action stage, this kind of classification is not clear. In addition, according to the degree of compression, some articles generally divided the processes in to bulk densification and surface densification.[21] Under this classification, the similarities and differences of the processes are not fully reflected.

In this paper, mechanical densification processes are classified according to the presence or absence of adding steam before or during compression. This classification system emphasizes two points. Firstly, the steam acting on the wood is provided by a steam generation system rather than water evaporation form the wood itself. Secondly, the steam discussed in that classification is used before or during compression. On the basis of the above two points, mechanical densification process can be divided into three categories: non-adding steam (NAS) densification, adding steam (AS) densification and others. According to the outstanding feature of the processes, above three kinds of processes are further subdivided. The classification of mechanical densification process in this paper is shown in the Fig. 1. and Table 1. Their processes and characteristics are summarized and discussed as follows.

NAS densification

The typical processing technique of densifying wood by NSA densification requires four steps: softening or plasticization, compression and cooling. The porous structure and the soft-enable property under heat of wood make it feasible to be densified, as shown in Fig. 2. During densification, wood is compressed by reducing the void space inside and between wood cells, with the help of softening treatments to prevent cell wall fracture during compression.[22,23]

NAS densification process

According to the compression degree of wood, NAS densification can be further classified into bulk and non-bulk densification. After bulk densification, wood is densified entirely in the compression direction and the density increases relatively evenly along the compression direction. While non-bulk densification makes only surface layer(s) or a certain inner layer of wood densified. Generally, the compression set (C, that is

Table 1. Mechanical densification process and corresponding main references.

Mechanical densification process		Main references
Non-adding steam (NAS) densification	Bulk densification	[6,24–26,28,69,72]
	Non-bulk densification	[32,33,41,46,52,54]
Adding steam (AS) densification	Adding steam at Pre-compression stage	[82–86]
	Adding steam at pre- and compression stages	[19,79,91–97]
Others		[100–107]

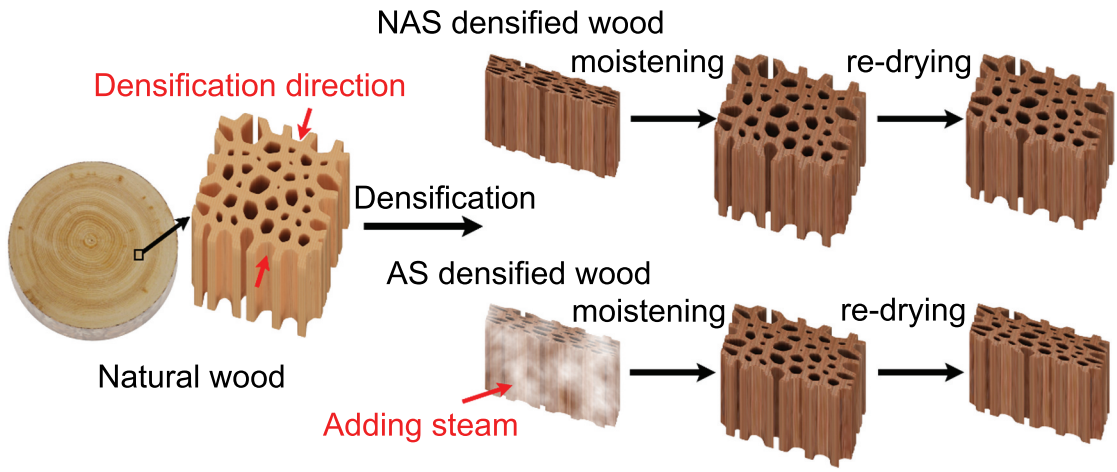

Figure 2. Densification and recovery of densified wood.

defined by equation 1) of bulk densification is higher than that of non-bulk densification. According to the tree species and application requirements, the compression set of bulk densification is usually between 40% and 60%,[6,24–29] while in non-bulk densification processes, the compression set is usually between 6% and 30%.[22,30–35]

$$C = \frac{T_0 - T_c}{T_0} \times 100\% \tag{1}$$

where, T_0 and T_C are the radial thickness of wood before and after compression, respectively.

2.1.1.1 Bulk densification. Equipment used for bulk densification is generally a hydraulic press machine (Fig. 3) which is mainly composed of top and bottom pressure platens, hydraulic system and control system. Before the densification process, samples are air-dried to reach a moisture content of around 12%.[28] Moreover, to ensure that plastic deformation occurred and to reduce defects caused by pressing, samples in some processes were controlled to reach an initial moisture content of 14–16%,[26] 25 ± 1%,[2,36] 30%[37] or even submerged in boiling water.[27] Then the samples are placed between the two pre-heated platens with a light pressure. After their interior temperature rises to preset temperature, samples are compressed

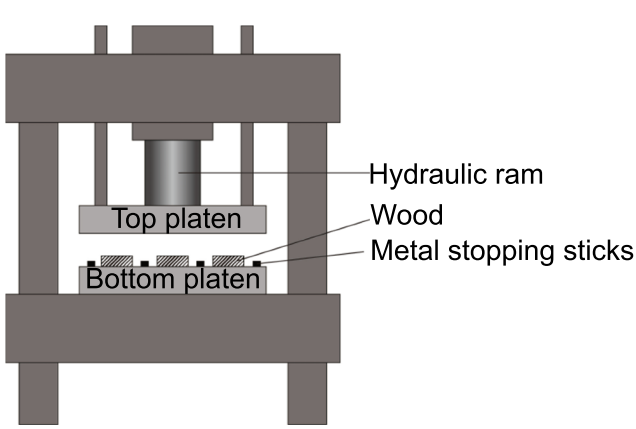

Figure 3. Schematic diagram of the hot press used for bulk densification.

to a target thickness, which is usually controlled by metal stopping sticks or computer program, and the platens are maintained in the same position for a while. Specifically, the preset temperature is required to be higher than the glass transition temperature (T_g) at which wood can transition from rigid state to a thermoplastic state,[38,39] so that wood can be reshaped by external load. The temperature used for compression varied from 100°C to 230°C according to the tree species, moisture content and the product requirement and the densification duration is also quite different in different studies. For example, Li et al.[24] compressed samples at 210, 220, and 230°C for 5, 12.5, and 20 minutes, and the whole system was then cooled until the temperature in the core of sample was below 60°C. In the study of Pelit et al.,[28] the densification process was done at 100 ± 5°C and 140 ± 5°C for 10 minutes. Some studies used much longer time. For example, in the study of Welzbacher et al.,[6] spruce (*Picea abies* Karst) samples were compressed at 140, 160, 180, 200°C for 30, 60, 120, 240 minutes. After compression, samples are removed from the press after the platens are cooled down to less than 100°C.[26–28] In particular, in order reduce crushing due to continuous deformation, the wood compression process can be divided into several stages. For example, Li et al.[37] compressed Scots pine (*Pinus sylvestris* L.) samples from 30 mm to 15 mm in three stages, each stage included that the sample was compressed by 5 mm and held for 300 seconds.

2.1.1.2 Non-bulk densification. For some applications, bulk densification is not essential and densification of surface or some certain parts is sufficient.[40] Compared with the bulk densification, non-bulk densification has some distinct advantages, such as shorter process and less wood volume loss.[22,33,34,41] Furthermore, intermittent pressing was used to compress wood in some processes to avoid crushing during compression, in which the wood compression process was divided into several cycles. Each cycle consists of a certain compression time and unloading time. After several cycles, the wood was compressed to the target thickness.[29,31,37] According to the position of the densified layer(s), non-bulk densification process can be divided into single-sided surface densification, double-sided surfaces densification, internal densification and center densification.

Single-sided surface densification means that only one side surface density of compressed wood increases significantly. The single-sided surface densification is usually achieved by controlling different temperature between top and bottom press platens.[32,33,41–45] An example of density profile is shown in Fig. 4. The equipment used for this type process is also the hydraulic press machine. Unlike the bulk densification machine, only one platen, which is usually the lower platen, would be heated. The sample is placed on the heated bottom platen. The top platen pressurizes the samples to be compressed until the target thickness is reached. After a period of holding, the bottom platen is cooled down to below 100°C before opening the press to diminish the immediate spring-back, and the single-sided surface densified wood is obtained.

Double-sided surfaces densification means that both surfaces of wood are compressed. The densities of the two compressed surface layers are significantly higher than that of the interior. Both platens of the hydraulic press machine are required to be heated, so that both upper and lower surfaces of wood samples are subjected to heating and softening. Short-term compression is an effective method to produce double-sided surfaces compressed wood. For example, in studies of Bekhta et al.,[46–48] woods were densified between heated platens for only 4 minutes. After being removed from the hot platens, samples were cooled down to about 25°C naturally. Double-sided surfaces densification also can be achieved by controlling the different moisture content between surface layers and interior of wood, because higher moisture content can make wood easier to be compressed. For example, Gao et al.[30,49] used this method to study the surface densification of poplar (*Populus tomentosa*). Paraffin was used to coat specimens on the radial and transverse surface, followed by 2-hour water soaking before densification treatment. Therefore, only the two tangential surfaces contacted with water directly and consequently had higher moisture content than internal part. After compression, the densities of the two surface layers increased significantly, while that of the core layer did not change substantially. Another method of controlling high surfaces moisture content is to spray water on the surfaces of wood. Zhan et al.[50] used this method to densify needle fir (*Abies nephrolepis*) surfaces. The samples were water-sprayed evenly on both surfaces to increase the moisture content, and then placed between the preheated upper and lower hot platens and compressed. Moreover, Sadatnezhad et al.[51] reported a continuous process with water spraying.

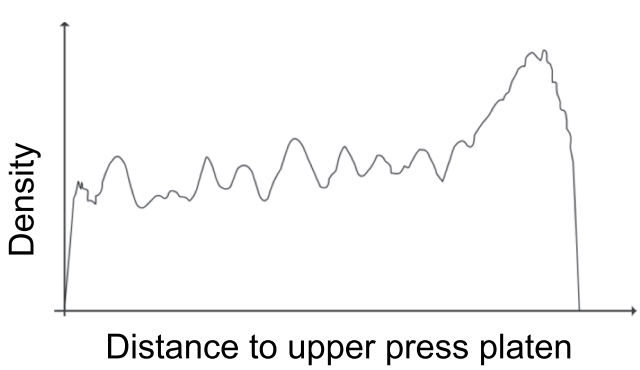

Figure 4. An example of density profile after single-sided surface densification.

Internal densification refers to having two compressed layers inside wood and center densification refers to having single compressed layer located at the center of wood. Internal or center densified wood can be gained by the sandwich compression process.[30,52–54] Wood samples were firstly immersed in water, and then preheated between two hot platens. One way was to control preheating time; another way was to control preheating temperature. After preheating, wood samples were compressed and cooled before being removed from press. It was found that when preheating time was short or temperature was lower, surface layers with high moisture content were firstly softened and compressed, and double-sided surfaces densified wood was produced. With the temperature rising and the preheating time extending, the moisture of surface layers evaporated gradually and the moisture content of interior became relatively higher, so that the location of densified layers moved from the surfaces to interior and finally converged to one at center (Fig. 5).[52]

Problems of NAS densification and the possible solutions

NAS densification can effectively compress wood, increase wood density and mechanical properties. But this kind of process has two problems that seriously affect product application and factory production. One is the poor dimension stability of densified wood, and another one is the long time and large energy consumption caused by cooling.

Dimension stability of densified wood. The dimension stability of compressed wood treated only by NAS densification is very poor.

On the one hand, single-sided densified wood has a tendency to develop cupping deformations. Belt et al.[22] explored this issue and considered that the asymmetrical distribution of density and the gradient moisture content due to single-sided heating were important causes of the cupping behavior. In addition, they thought that the gradient moisture content due to single-sided heating was the most important factor. The schematic diagram of cupping occurred on the undensified side and the densified side were shown in Fig. 6.

On the other hand, almost all NAS densified woods have serious shape recovery problems, including instantaneous spring-back after opening the press and deformation recovery when exposed to high humidity or directly soaked in water (Fig. 2). After removing the press pressure, the instantaneous spring-back (about 5%[26]) of densified wood will happen, and the spring-back even reach between 15% to 20% after conditioning.[6,28] If the densified wood is exposed to high

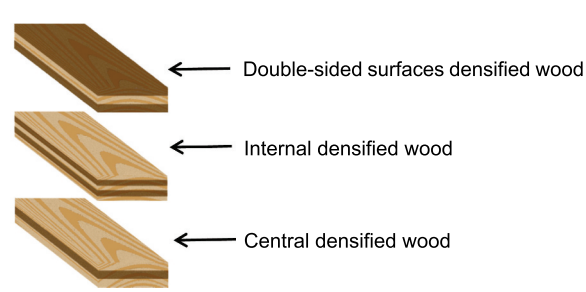

Figure 5. Schematic diagram of sandwich compressed wood.

Figure 6. (a) Cupping on the undensified side; (b) cupping on the densified side.

humidity (85% relative humidity) for a long time, the deformation recovery will be about 20% to 30%. Further, if it is immersed in water, the recovery rate can be as high as more than 60%,[6,26,55] or even 90%.[28] As we all known, dimension stability is of great significance for the application of densified wood. To minimize the compression recovery and spring-back, Norimoto et al.[56] proposed that reduced recovery can be achieved by relaxing internal stressed stored in the microfibrils and matrix, by reducing hydrophilic cell wall constituents, and by forming new cross-linkages between matrix molecules. Based on the above theory, many researchers have carried out new explorations. After summarizing the processes, there are two ways to solve above problems based on the NAS densification process: post-treatment and compression in a closed press.

Three types of post-treatment, air-heat treatment, oil-heat treatment and steam treatment, can be used to reduce spring-back and recovery. Among them, the air-heat post-treatment is a relatively simple method. This method can improve the dimension stability of densified wood by the facts including degradation of hygroscopic hemicellulose polymers,[57,58] and cross-linking of hydroxyl groups between adjacent cellulose chains or aromatic rings in lignin.[59,60] Furthermore, air-heat post-treatment can release internal stress generated during compression.[25] Pelit et al.[28] analyzed the effect of air-heating on swelling and spring-back of NAS bulk densified wood. The oven dried densified wood samples were air-heat treated at 212°C for 2 hours. The findings showed that the dimension stability of densified wood was obviously improved. The spring-back and deformation recovery of densified samples were decreased by up 88% and 85%, respectively.

Oil-heat treatment is also a commonly used post-treatment to fix the densified wood due to the effective performance of oil as a transfer media of heat. Dubey et al.[26] post-treated the compressed pine (*Pinus radiata*) samples with hot linseed oil at 160–210°C in a bath. This study demonstrated that the spring-back and deformation recovery percentages of samples treated with oil-heat post-treatment decreased with the increase of oil-heat temperature. The recovery caused by high humidity and immersion could be reduced by up to 84.4% and 91.5%, respectively. The improved dimension stability can be explained by the fact that oil-heat treatment can lead to further degradation, mainly the hemicelluloses degradation, and can relieve the residual stress buildup during the compression more easily. They also found that oil-heat treatment could improve samples' biological resistance. However, the main drawbacks of air-heat or oil-heat treated wood is the reduced mechanical properties.[6,26,35,61] But the performances of treated wood still superior to natural wood, and can meet the needs of most applications.

Compared with the above treatments, steam post-treatment, which is used widely, can not only improve significantly the compressed wood dimension stability, but also can minimize discoloration phenomenon and performances degradation.[62] Laine et al.[55] reported the effect of steam post-treatment on the deformation recovery. The surface-densified pine woods were treated in non-pressurized oven by heating followed by steam injection at 200°C for 3 hours. After post-treatment, the dimension stability was tested by several soaking-drying cycles, which showed that the hydrothermal post-treatment considerably reduced the recovery, to a value at least 74% lower. Gao et al.[49] also found that atmospheric and pressurized steam post-treatment reduced the average recovery of surface-densified polar significantly from 12.9% to 4.1% and 1.5% respectively. The improved dimension stability of densified wood profiting from steam post-treatment is most likely due to the fact that the lignin-hemicellulose matrix is softened, that allows internal stress to relax[55] and microfibrils to move causing an increased cellulose crystallinity.[63]

In summary, reasonable post-treatment methods can effectively make up for the shortcomings of NAS compressed wood. However, the post-treatment process usually takes several hours, and consumes more energy. In addition to the above post-treatment methods, resin impregnation and chemical treatment can also be alternative.[64–68] But these methods not only cost much, but also reduce the recovery and even pollute the environment.

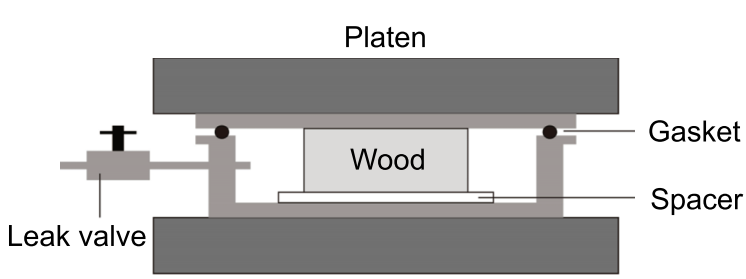

Figure 7. Compression in a closed press.

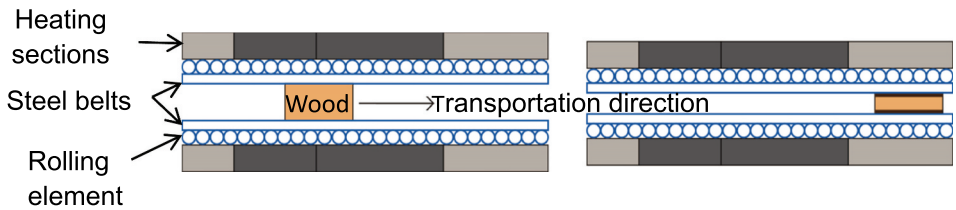

Figure 8. Schematic diagram of the continuous densification process.

Besides post-treatment, the dimension stability of NAS densified wood can also be improved by compression in a closed press. The hot press for this type process is generally equipped with a metal frame which has a gasket making the enclosure airtight as shown in Fig. 7.[69–71] Under the heating of hot pressure platens, the steam is converted from the water inside samples and fills in the closed environment. In order to using moisture inside the wood itself, samples are usually soaked in water for having high moisture content. After compression, the pressurized vapor is released by opening the leak valve. Many researchers have reported that the closed system with steam was beneficial to the shape fixation of densified wood. For example, Fukuta et al.[72] conducted soaking-boiling -drying tests on 50% densified Japanese cedar (*Cryptomeria japonica*) and found that the recovery was almost 0% at 180°C for 15 minutes in closed system, while the recovery became almost 0% when the wood was treated for 8 hours by the open system at 180°C.Therefore, compared with NAS densification in open system, the closed method is more effective for deformation fixation. In addition, Yi et al.[73] used similar equipment to densify Japanese cedar, and the results showed that this closed system could significantly increase the dimension stability of compressed wood with improved mechanical properties. When the wood was densified at 200°C for 5 minutes or at 180°C for 13 minutes, the deformation was almost constant. However, the compressed wood was easy to bulge or rupture because of the high vapor pressure inside wood. Therefore, this method must have a cooling stage before opening the press platens to reduce the pressure. In order to facilitate the release of vapor inside compressed wood, Fukuta et al.[74] adopted a method of drilling holes in the wood. However, this operation reduce the mechanical strength of the material itself.[75]

2.1.2.2 Long time and large energy consumption. Another characteristic of NAS densification is that the cooling stage is required. This is because the moisture in the wood would become steam under the action of high temperature, resulting in a high vapor pressure inside wood. Even if the wood compression process is conducted at an open system, the vapor pressure cannot be reduced sufficiently if the wood sample is large. Once the press platens are opened, the compressed wood is prone to bulge or burst, and explosion even happened before the opening of press platens. Therefore, it is necessary to cool the compressed wood at the last stage of densification process in order to avoid the destructive effect of vapor pressure. There are two common cooling methods. One is natural cooling, which usually takes long time. Another is by the water-cooling system. Although this method can decrease temperature quickly, it takes lots of water. Additionally, some studies not only had cooling stage, but also used perforated mental platens to enable steam evaporation.[44,76,77]

Furthermore, before compressing the next batch, the platens must be heated again, which usually consumes large amount of energy and time. To reduce the time and energy consumption caused by reheating, the continuous densification process reported by Sadatnezhad et al.[51] can give us some inspirations. This machine showed as Fig. 8 consists of four adjustable heating sections, two set of rolling elements, and movable steel belt. As it is transported through the press, the wood can be softened and compressed and finally cooled to room temperature by different units. By this way, the densified wood can be produced continuously.

AS densification

AS densification employs a combination of heat, pressure and additional steam. In particular, the additional steam in AS densification process doesn´t refers the steam only used during post-treatment or converted from the water inside wood as mentioned in section.

Additional steam treatment in AS densification process can soften wood effectively

It is because that increasing wood moisture content can decrease T_g, and consequently wood can be softened at a lower temperature.[78] Another reason is that steam injection can lead to a quick temperature rise in the wood interior due to the quick heat transfer caused by vapor convection.[79]

Moreover, compared with NAS densification, AS densification not only can improve mechanical properties, but also shows great advantages in reducing spring-back and deformation recovery of compressed wood,[19,79,80] as shown in Fig. 2. There have been some theories that explain the mechanism of shape fixation of AS densification. One of theories is the internal stress can be released under the action of steam and heat. Obataya et al.[69] proposed that the irreversible shape recovery of densified wood exposed to moisture was caused by two reasons. One reason is that the elastic energy stored in fiber is released when the amorphous matrix is softened. Another is the inflation of compressed cells with swollen cell wall due to water uptake, which likes the inflation of folded swimming ring due to air uptake. By using steam, the elastic stress, the major driving force for shape recovery, was relaxed. Therefore, compressed wood only showed reversible changes generated from the swimming ring-like shape change of cell lumen. It was also widely reported that hot steam can cause degradation of hygroscopic cell wall components, such as hemicelluloses, thereby improving dimension stability of compressed wood.[4,81] In addition, Chen et al.[70] suggested that recrystallization of cellulose was the most reasonable mechanism involved in the shape fixation due to steaming. In their study, both the

crystallinity index and crystal width both increased with steam treatment, corresponding to compression recovery decreased significantly.

AS densification process

Similar to NAS, AS densification process can produce both bulk and non-bulk densified wood. Since compression with adding steam is the most prominent feature of AS densification, this paper will classify the AS densification process into two categories according to the working stage of adding steam. One is at pre-compression stage, and the other is both at pre- and compression stages.

Adding steam at pre-compression stage. Using adding steam at pre-compression stage is an effective method of softening wood. For example, in the study of Paril et al.,[82] the plasticization stage of wood was carried out in an oven which provide saturated steam for 4 hours at 100°C. Then the softened wood was transferred to the hydraulic press for compression. In the study of Inoue et al.,[83] the specimens were steamed in an autoclave by introducing high pressure steam at temperature form 120°C to 220°C for 10 minutes. Pre-steamed specimens were compressed by hot pressing to 50% of their initial thickness. And they found that shape recovery after soaking in boiling water decreased dramatically when the pre-steaming temperature exceeded 170°C, and the deformation was almost completely fixed by pre-steaming at 220°C for 10 minutes, which was due to the reduction of accessibility to moisture and stresses stored by deformation. However, this kind of pre-treatment would greatly increase the processing time.

Besides the above processes, the viscoelastic thermal compression (VTC) is also a typical AS densification process. Equipment of VTC process is a closed heat press that integrates dynamic heat, steam, and mechanical compression schedule. The apparatus is mainly comprised of lid, bellows and internal press (Fig. 9). The internal press is used for supporting and compressing wood, and can be controlled for heating and cooling independently.[84] In this process, wood is softened by high temperature and saturated steam. After the wood is sufficiently softened, the steam pressure is released. Then the wood is compressed immediately, and platens are heated to a higher temperature. In the last step, wood is cooled under compression.[85–88] Because wood is in a hygro-thermal closed environment, the wood can be softened at lower temperature, which can save energy and avoid the negative effect of excessive temperature on wood mechanical properties. In addition, the whole process of VTC can be completed in about 20 minutes, which makes the production process efficient. Kutnar et al.[78,89,90] found that the modulus of elasticity (MOE) and modulus of rupture (MOR) of wood were improved by 37%-129% and 32%-102% after 37.5%-58.3% densification by VTC treatment, and the deformation recovery after five soaking-drying cycles was only about 8%. And due to the short time and moderate temperature of VTC, the loss of wood quality was small. In addition, the microscopic observation revealed VTC can densify the wood on the premise of ensuring that the cell wall is not destroyed.

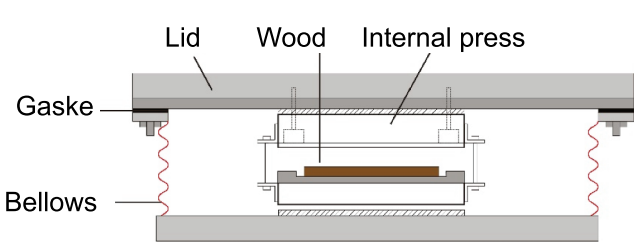

Figure 9. Schematic diagram of the VTC device.

Adding steam both at pre- and compression stages. There have been many studies on AS densification which adding steam works both at pre- and compression stages. According to the perspective of the spatial characteristics of steam treatment, they can be divided into two categories: one is in closed system, and the other is in open system.

Ito et al.[91] developed a sealed compressive-molding device with a press cylinder, in which wood can be compressed combining with high-pressure steam. Sugi (*Cryptomeria japonica* D. Don) specimen was softened by saturated steam at 150°C for 3 minutes, and then compressed to 33.3% of the initial thickness in the closed mold, followed by holding for 2 minutes at 190°C or 200°C to fix shape change.Navi et al.[92] also developed a AS densification equipment with a hydro-thermal chamber. The specimen placed in the mold was compressed by more than 65% in this hydro-thermal chamber. Heger et al.[15] used similar closed apparatus to densify wood. Specimens were plasticized at 140°C for 10 minutes and compressed by 66% under the condition of saturated steam in the reactor. After post-treatment at 140°C-200°C, the steam was vented out of the treating chamber to purge the reactor, and specimens were immediately cooled below 100°C and removed from the reactor. One advantage of above equipment is that they combine softening, compression and fixation process, which improves wood mechanical properties and dimension stability obviously. And the specimens can be heated and softened rapidly by steam without addition press heating. As the size of compression mold mentioned above is small, only small-size wood can be densified. For example, the size of sample compressed with the device developed by Navi et al.[92] was only 40 × 30 × 30 mm. The practical application of such small-sized densified wood is very limited.

Besides the above processes in a closed system, some AS densification processes allow wood to be compressed in an open environment.

In 2012, we have proposed a densification process that integrates steam pre-treatment, hot pressing with steam and steam post-treatment.[93] The equipment shown as Fig. 10 was used. On both upper and lower press platens holes are distributed evenly, which are used for injecting and venting steam. Therefore, the device can not only allow the steam to be evenly applied to the wood but also avoid the damage of compressed material caused by the excessive internal vapor pressure. Specimens are softened by the pre-heated platens and injected steam, followed by compressing to a target thickness with steam injecting. Once the specimens reach the target thickness, the distance between the two platens is held for a preset period with continuing steam injection. At the end, before platens opening, steam injection is stopped, and the steam and vapor are vented by vacuuming through the holes (Fig. 11). Our

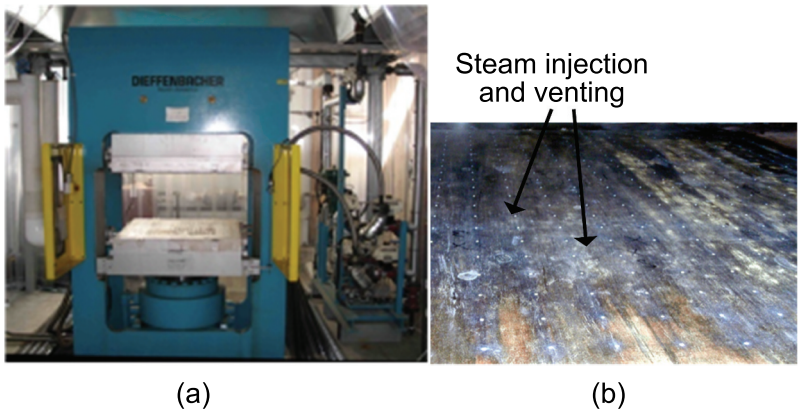

Figure 10. (a) Hot press machine for AS densification; (b) platen with evenly distributed holes.

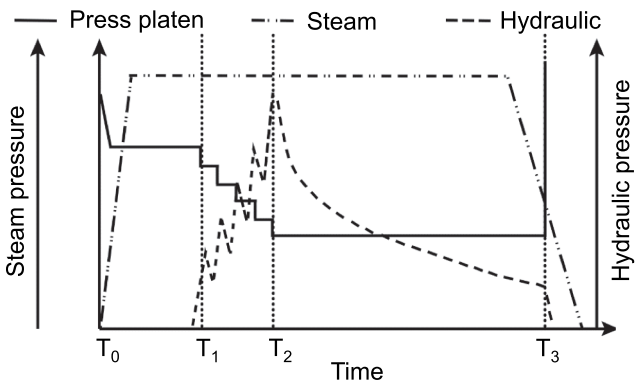

Figure 11. Schematic diagram of AS process. T_0-T_1: pre-treatment; T_1-T_2:compression; T_2-T_3:post-treatment.

previous studies showed that with the increase of temperature, the hygroscopicity and dimension stability of compressed wood decreased and increased, respectively. When the temperature reached 220°C, there was almost no compression set recovery caused by moisture absorption. In addition, this process was relatively simple with short duration. One reason is that all the treatments are completed continuously, which greatly improves production efficiency. Another reason is that the hot press platens do not need to be cooled down for avoiding bulge and explosion, which can greatly reduce the energy consumption and time caused by the re-heating for next densification batch.

Moreover, this AS process has been applied in many densified wood studies. For example, Fu et al.[79,94] used the above AS process to investigate the performances of densified maple wood (*Acer saccharum Marsh*). Results suggested that, at the same temperature, the mechanical properties and dimension stability of the wood densified with steam were higher than those of the wood densified without steam. For example, the MOR and MOE of specimens densified with steam could increase by up to 38.8% and 53.1%, respectively. Koumba et al.[95] and Cruz et al.[96] also used this AS process when they investigated the chemical components change and densified wood properties. On the basis of the above AS densification process, we also carried out oil heat post-treatment on the compressed wood at 180°C-220°C for 1–3 hours.-[93,97,98] After post-treatment, the dimension stability of compressed wood was furtherly enhanced, and the recovery could even reduce to 0%.[93]

However, the holes for injecting and venting steam on the hot pressing platens are easy to cause protrusions on compressed wood under high pressure during densification.[5] The protrusions block to some extent the infection and venting of steam, which causes that steam cannot distribute uniformly on the compressed wood. In order to solve this problem, we used sintered metal mesh laminate (SMML) as the isolation layer between wood and hot pressing platens.[99] The permeable porous SMML makes it possible for steam to distribute evenly on compressed wood and provides space for high-pressure vapor to release from densified wood. Due to the fine smooth surface of SMML, the densified wood obtained with this method showed smooth surface with uniform color.

Other Methods

High-pressure densification

The high-pressure (HP) densification technology is a non-thermal process. In HP treatment, the wood sample is sealed in a vacuum polymer bag, then placed in the HP chamber for densification. The chamber is filled with pure water. Wood is densified by the extremely high isostatic pressure (50 MPa-200 MPa) generated by water. The whole process is carried out at room temperature.[100–102] The process time is very short (only 5 minutes or even 30 seconds), and a much higher isostatic hydrostatic pressure can be employed. Moreover, there is no need for thermal treatment. With this method the part with low strength is preferentially compressed, so that the densified product is irregularly shaped.

Densification process with microwave plasticization

Plasticization is an essential step in wood densification process. Microwave technology as one of the methods for wood plasticization has several advantages compared with traditional methods, such as short duration, high energy efficiency. That is because the molecules of wood can directly absorb the energy provided by microwave allowing for an internal heating process quickly.[103] Before entering the microwave modification process, the samples are soaked in distilled water. Then samples are placed on the conveyor belt and pass through the modified chamber at a certain conveying speed and microwave power. The plasticized wood is compressed by hot pressing

platens.[104] Döemény et al.[103,105] reported that wood plasticization with microwave could be considered as an effective approach and has application potential in the wood flooring industry. However, only few reports were found on wood densification with the help of microwave.

Densification process with removing lignin/hemicellulose

In 2018, Song et al.[106] proposed a wood densification process with two steps. The most remarkable characteristic of this process is that the low density (0.43 g/cm^3) wood can be completely densified and very high strength material with extremely high density (1.30 g/cm^3) can be obtained. And in the recent study of Shi et al.,[107] wood density was increased by 169% by this process. The process is divided into two steps. Firstly, mostly of the hemicellulose and lignin are chemically removed, which makes wood more porous and less rigid. Secondly, upon hot-pressing, the porous cell walls and lumina collapsed entirely. Song et al.[106] found that with the help of removing lignin and hemicellulose wood could be densified completely, and the mechanical strength obtained was not only much higher than that of undensified wood, but also superior to that of densified woods obtained by other densification processes. The main reason was that the chemical solution removed lignin and hemicellulose, which increased the proportion of cellulose content. After compression, the interfacial area among nanofibers increased and a large number of hydrogen bonds were formed. Therefore, much more energy was needed to fracture this kind of densified wood. However, for a small-size sample, the two-step process took very long time (more than 24 hours) and the chemicals used are not environmental-friendly.

Mechanical properties of densified wood

The main purpose of mechanical densification is to improve the mechanical properties of low or medium-strength woods, so as to expand the application field of fast-growing wood. For example, in the study of Li et al.,[24] the Brinell surface hardness and bond strength of densified balsam fir (*Abies balsamea* (L.) Mill.) compressed by 60% were about two times larger and 30% higher than that of undensified fir. In the study of Pertuzzatii et al.,[27] after 40% compression, the density of pine (*Pinus elliottii*) increased from 0.59 g/cm^3 to 0.99 g/cm^3, which increased MOE and MOR by 85% and 294%, respectively. And in their study, the density and mechanical properties of eucalyptus (*Eucalyptus grandis* Hill) were also improved significantly after densification. At present, due to the excellent mechanical properties of densification wood, it has been widely used in decoration, construction, furniture and other fields, and even replaced high-strength wood materials. The rapid development of densification technology and the application of densified wood products are of great significance not only to solve the problem of imbalance between supply and demand of wood materials, but also to the promotion of environmentally friendly materials and national economic development.

There are many factors that affect the mechanical properties of densified wood materials. In this paper only several typical examples are summarized, and some key process factors affecting the properties are analyzed as follows.

Among all the influencing factors, tree species and compression parameters (including compression degree, temperature and time) are dominant. Under the same densification conditions, there are performance differences between different tree species. Pelit et al.[25] used NAS bulk densification process to densify Uludağ fir (*Abies bornmulleriana* Mattf.), linden (*Tiliagrandifolia* Ehrh.) and black poplar (*Populusnigra* L.). After densification at the same compression degree (50%) and compression temperature (100°C), MOE of fir, linden and poplar increased by 39%, 36% and 44% respectively, and densified poplar had the highest level of MOE. Similar results were obtained in the tests of MOR and compressive strength parallel to the grain (CS). In addition, the mechanical tests showed that MOE, MOR and CS of densified samples were increased with compression degree and decreased with compression temperature. An unsuitable treatment temperature can chemically degrade wood cell wall, and lead to loss of strength and greater brittleness.[9] Similar conclusions were also found in our research,[19] after AS densification, the Brinell hardness of aspen (*Populus tremuloides*) and poplar were increased by one and two times because of the dense of wood structure. However, when the hot-pressing temperature was higher than 180°C, the hardness decreased obviously, which was explained by the degradation of hemicellulose and lignin.

Post-treatment performed to provide dimension stability can also significantly affect the mechanical strength of densified wood. The reduced strength (especially MOR) of compressed wood after air-heat post-treatment was shown in the study of Pelit et al.[25] and they obtained similar results in the study of the Brinell hardness of densified Scots pine and Eastern beech (*Fagus orientails* L.).[42] Difference results have been reported in some studies. For example, Kubojima et al.[108] reported that short-term heat treatment could increase the mechanical properties of Sitka spruce (*Picea sitchensis* Carr.), while properties decreased when heat treatment over 4 hours. Shi et al.[109] reported a 6% increase of MOR of birch (Betula spp.) after heat treatment, but 49% decrease of heat-treated spruce (*Picea* spp.) was observed. Therefore, the effect of heat post-treatment on the mechanical properties of wood depends on wood species and treatment condition including time and temperature. In our study,[93] because of the degradation of polymer matrix, Brinell hardness and maximum tensile strength of densified aspen were negatively correlation with oil-heat treatment temperature and duration time. Contrary, MOE slightly increased after oil-heat treatment due to the increase in the relative amount of crystalline cellulose. On the whole, the mechanical properties of post-treated densified wood were much higher than those of non-densified wood, which benefited from the mechanical densification.

Concluding remarks and perspectives

Wood mechanical densification is a promising technology for improving wood properties, and consequently providing great potential to expand wood application domain and develop new products. In this paper, wood mechanical densification processes were summarized and classified into NAS densification, AS densification and other methods, and they were further

classified according to the degree of densification or the work stage of adding steam. Moreover, the technological process, advantages and disadvantages, process improvement, product performances of each densification process were described and discussed. By untangling above knowledge, we can see that wood mechanical densification processes have been well developed.

However, there are still some drawbacks in these methods that need to be further improved. The following points can be considered for improvement. (1) More effective fixation method. Although some processes have reduced the shape recovery of densified wood, it is still a problem for some applications, which require high dimension stability. (2) Reducing or avoiding negative effects of high temperature, for example, embrittlement, rapid aging, and dark color, especially in NAS densification processes, and gluing. (3) Reducing energy and water consumption. Wood mechanical densification processes require a lot of energy for heating, and steam treatment needs water additionally. (4) Improving production efficiency and realizing continuous production for the further application in industry. (5) Life cycle consideration. The whole chain from nature material to end-of-life is needed to be considered and optimized. Overcoming these challenges requires cooperation between entrepreneurs and researchers from multiple fields. With the improved wood densification technologies, wood resources can be used in broader fields with higher added value.

Acknowledgments

The authors acknowledge the financial support from the Foundation of International Centre for Bamboo and Rattan (1632019002) and the Natural Science Foundation of China (32071856)

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the [National Natural Science Foundation of China 32071856]; [Foundation of International Centre for Bamboo and Rattan 1632019002].

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